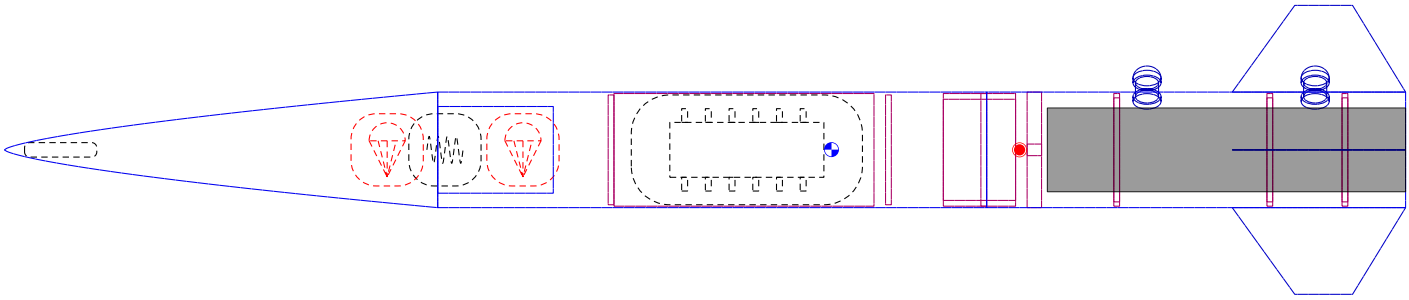
Rocket Design



Rocket

Stages: 1

Mass (with motor): 694 g

Stability: 1.63 cal / 13.4 %

CG: 28.6 cm

CP: 35.1 cm

G80T-4

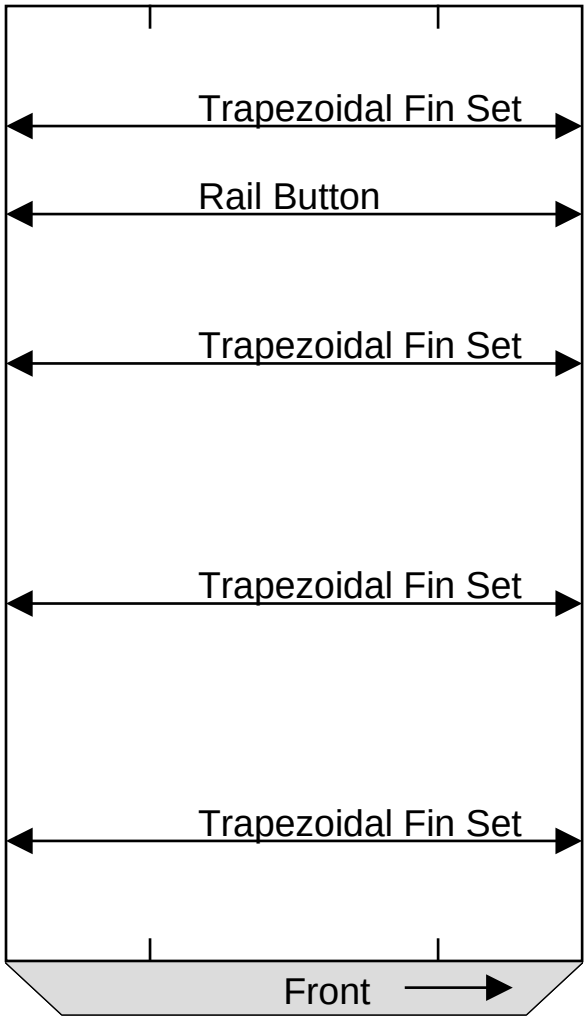
Altitude	1057 m
Flight Time	123 s
Time to Apogee	14.1 s
Optimum Delay	12.3 s
Velocity off Pad	16.2 m/s
Max Velocity	171 m/s
Velocity at Deployment	11.6 m/s
Landing Velocity	7.03 m/s

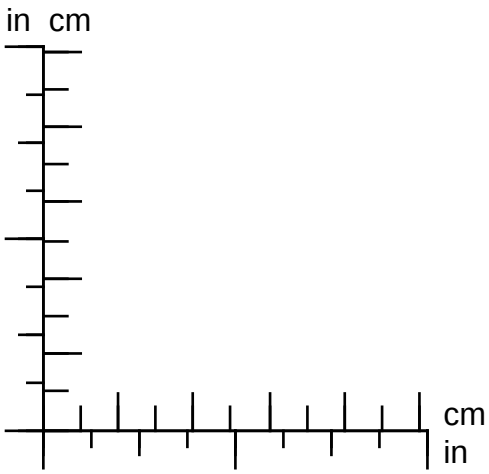
Motor	Avg Thrust	Burn Time	Max Thrust	Total Impulse	Thrust to Wt	Motor Wt	Size
G80T	78 N	1.71 s	102 N	133 Ns	11.46:1	62.5 g	29/124 mm

Parts Detail

Stage: Sustainer

	Nose Cone	Fiberglass (1.85 g/cm³)	Power series	Len: 15 cm	Mass: 56.6 g
	Parachute	Ripstop nylon (67 g/m²)	Dia_{out} 35 cm	Len: 2.5 cm	Mass: 12.1 g
	Shroud Lines	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)	Lines: 6	Len: 52 cm	
	Shock Cord	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)		Len: 65 cm	Mass: 1.17 g
	Aluminium Tip		Dia_{out} 0.5 cm		Mass: 25 g
	Avionics Bay and Main Parachute	Fiberglass (1.85 g/cm³)	Dia_{in} 3.6 cm Dia_{out} 4 cm	Len: 19 cm	Mass: 83.9 g
	Mass Component		Dia_{out} 3.8 cm		Mass: 245 g
	Parachute	Ripstop nylon (67 g/m²)	Dia_{out} 45 cm	Len: 2.5 cm	Mass: 18.2 g
	Shroud Lines	Elastic cord (round 2 mm, 1/16 in) (1.8 g/m)	Lines: 6	Len: 70 cm	
	Tube Coupler	Carbon fiber (1.78 g/cm³)	Dia_{in} 3.5 cm Dia_{out} 3.9 cm	Len: 2.5 cm	Mass: 10.3 g
	Bulkhead	Cardboard (0.68 g/cm³)	Dia_{out} 3.9 cm	Len: 0.2 cm	Mass: 1.62 g
	Inner Tube	Fiberglass (1.85 g/cm³)	Dia_{in} 3.8 cm Dia_{out} 3.9 cm	Len: 9 cm	Mass: 10.1 g
	Bulkhead	Cardboard (0.68 g/cm³)	Dia_{out} 3.8 cm	Len: 0.2 cm	Mass: 1.54 g
	Bulkhead	Cardboard (0.68 g/cm³)	Dia_{out} 3.8 cm	Len: 0.2 cm	Mass: 1.54 g
	Motor and Fin Mount	Fiberglass (1.85 g/cm³)	Dia_{in} 3.6 cm Dia_{out} 4 cm	Len: 14.5 cm	Mass: 64 g
	Trapezoidal Fin Set (4)	Fiberglass (1.85 g/cm³)	Thick: 0.25 cm		Mass: 18.9 g
	Engine Block	Carbon fiber (1.78 g/cm³)	Dia_{in} 0.4 cm Dia_{out} 4 cm	Len: 0.5 cm	Mass: 11.1 g
	Centering Ring (motor)	Carbon fiber (1.78 g/cm³)	Dia_{in} 3.6 cm Dia_{out} 3.9 cm	Len: 0.2 cm	Mass: 0.629 g
	Fin holder	Carbon fiber (1.78 g/cm³)	Dia_{in} 3.6 cm Dia_{out} 3.9 cm	Len: 0.2 cm	Mass: 0.629 g
	Centering Ring (Motor)	Carbon fiber (1.78 g/cm³)	Dia_{in} 3.6 cm Dia_{out} 3.9 cm	Len: 0.2 cm	Mass: 0.629 g
	Rail Button	Delrin (1.42 g/cm³)		Len: 0 cm	Mass: 2.35 g


Trapezoidal Fin Set
Rail Button
Trapezoidal Fin Set
Trapezoidal Fin Set
Trapezoidal Fin Set
Front


in cm
cm
in

